

Data analytics for personalized genomics and precision medicine**Lecture4 scribing**

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Lecture Outcome

- More about Dynamic Programming
- Gene expression matrix
- Introduction to sequence assembly and sequence mapping

1. More about Dynamic Programming**1.1 Recap from last lecture**

- Why dynamic programming? → to find optimal sequence alignment
- What is dynamic programming? → breaking down complicated problem into sub-problem and solve them recursively
- How to construct a dynamic programming table? → fill in cumulative gap penalties and the score for each column

1.2 Steps for dynamic programming

1. Define a scoring matrix

Scoring matrix:

	A	C	G	T
A	2	-7	-5	-7
C	-7	2	-7	-5
G	-5	-7	2	-7
T	-7	-5	-7	2

Gap penalty = -10

2. Fill in the dynamic programming table with arrows to preserve the path

		A	C	C	G	
		0	-10	-20	-30	-40
A	-10	2	-8	-18	-28	
C	-20	-8	4	-6	-16	
G	-30	-18	-6	-3	-4	

ACCG

ACG

$F(4,3) = -4$

3. The last cell ($F(4,3)$) represents the best alignment score
4. Trace back the optimal alignments (the blue arrows below)

		A	C	C	G
	0	-10	-20	-30	-40
A	-10	2	-8	-18	-28
C	-20	-8	4	-6	-16
G	-30	-18	-6	-3	-4

5. For sequences ACCG and ACG, the two optimal alignments is

- ACCG A_CG
- ACCG AC_G

1.3 Global and local alignments

- Global alignment: the last cell in the table ($F(4,3)$ in the above example)

- Local alignment: only part of the sequence (e.g. the most optimal alignment in the above example is AC AC which has the largest score 4)

1.4 More about scoring matrix

- Mutation → Mismatch
- Insertion/deletion, gene duplication → Gap
- Different scoring matrices are built from different databases (e.g. scoring matrix for protein sequence alignment below)

BLOcks Substitution Matrix (BLOSUM)

	A	R	N	D	C	Q	E	G	H	I	L	K	M	F	P	S	T	W	Y	V
A	7	-3	-3	-3	-1	-2	-2	0	-3	-3	-3	-1	-2	-4	-1	2	0	-5	-4	-1
R	-3	9	-1	-3	-6	1	-1	-4	0	-5	-4	3	-3	-5	-3	-2	-2	-5	-4	-4
N	-3	-1	9	2	-5	0	-1	-1	1	-6	-6	0	-4	-6	-4	1	0	-7	-4	-5
D	-3	-3	2	10	-7	-1	2	-3	-2	-7	-7	-2	-6	-6	-3	-1	-2	-8	-6	-6
C	-1	-6	-5	-7	13	-5	-7	-6	-7	-2	-3	-6	-3	-4	-6	-2	-2	-5	-5	-2
Q	-2	1	0	-1	-5	9	3	-4	1	-5	-4	2	-1	-5	-3	-1	-1	-4	-3	-4
E	-2	-1	-1	2	-7	3	8	-4	0	-6	-6	1	-4	-6	-2	-1	-2	-6	-5	-4
G	0	-4	-1	-3	-6	-4	-4	9	-4	-7	-7	-3	-5	-6	-5	-1	-3	-6	-6	-6
H	-3	0	1	-2	-7	1	0	-4	12	-6	-5	-1	-4	-2	-4	-2	-3	-4	3	-5
I	-3	-5	-6	-7	-2	-5	-6	-7	-6	7	2	-5	2	-1	-5	-4	-2	-5	-3	4
L	-3	-4	-6	-7	-3	-4	-6	-7	-5	2	6	-4	3	0	-5	-4	-3	-4	-2	1
K	-1	3	0	-2	-6	2	1	-3	-1	-5	-4	8	-3	-5	-2	-1	-1	-6	-4	-4
M	-2	-3	-4	-6	-3	-1	-4	-5	-4	2	3	-3	9	0	-4	-3	-1	-3	-3	1
F	-4	-5	-6	-6	-4	-5	-6	-6	-2	-1	0	-5	0	10	-6	-4	-4	0	4	-2
P	-1	-3	-4	-3	-6	-3	-2	-5	-4	-5	-5	-2	-4	-6	12	-2	-3	-7	-6	-4
S	2	-2	1	-1	-2	-1	-1	-1	-2	-4	-4	-1	-3	-4	-2	7	2	-6	-3	-3
T	0	-2	0	-2	-2	-1	-2	-3	-3	-2	-3	-1	-1	-4	-3	2	8	-5	-3	0
W	-5	-5	-7	-8	-5	-4	-6	-6	-4	-5	-4	-6	-3	0	-7	-6	-5	16	3	-5
Y	-4	-4	-4	-6	-5	-3	-5	-6	3	-3	-2	-4	-3	4	-6	-3	-3	3	11	-3
V	-1	-4	-5	-6	-2	-4	-4	-6	-5	4	1	-4	1	-2	-4	-3	0	-5	-3	7

- Scoring matrix is constructed after we search from the data base

2. Gene expression matrix

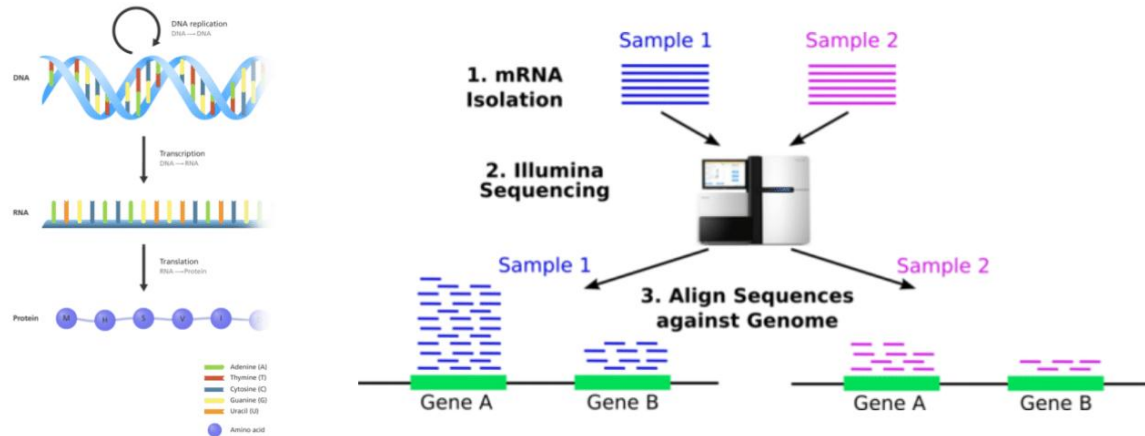
2.1 Why sequence data

- Central dogma
- Hidden genetic information
- Phenotype = Genotype (determined by sequence) + environment

2.2 Challenge that we face

- Highly similar genome among humans (genetic variation only account for 0.001% in genome)
- Small portion of genome encodes protein
- Knowing the sequence is not enough as gene expression difference may account for phenotype difference (same gene on different cell types could be variant)

2.3 How to get the gene expression matrix (from sequence data to data matrix)

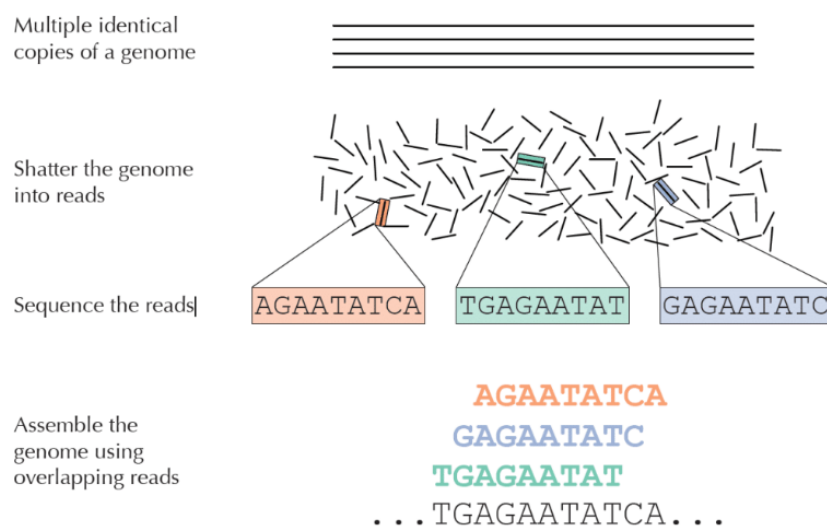


- Sequence RNA (abundance of mRNA)
- DNA → RNA (after transcription) → Amino acid (after translation)
- mRNA isolation → illumina sequencing → align sequence against genome
- E.g. Single-cell RNA-Seq (tissue → isolation and sequencing individual cell → read counts for each gene → compare gene expression profiles of single cells)

3. Introduction to sequence assembly and sequence mapping

3.1 Genome assembly

- Bases on short reads and overlap regions



3.2 Sequence mapping

- By sliding each read along the genome and calculate the difference
- E.g. sliding CCA across TAATGCCATGGATG

T	A	A	T	G	C	C	A	T	G	G	A	T	G
					C	C	A						
2	3	3	3	3	2	0	2	3	3	2	3	3	

Useful links:

Webserver for sequence alignment: https://www.ebi.ac.uk/Tools/psa/emboss_needle/

Biopython: <https://biopython.org/>